Universität Zürich[UZH]

Department of Informatics
Dynamic and Distributed Information Systems Group

Collaborative Intent Recognition in Political Debates using LLMs

Master's Project Description

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Introduction

Persuasion in online debates depends not only on argument content, but also on interaction patterns and the structure of threaded conversations (Tan *et al.* 2016; Öcal, Xiao, and Park 2021). Reddit's r/ChangeMyView (CMV) provides a useful setting for studying argumentative interaction, including many socially and politically contested discussions. However, short replies in such discussions are often context-dependent, making communicative intent difficult to identify.

Short replies may convey meaning indirectly through context, sentiment, and affective tone rather than explicit wording. Cues such as sarcasm, hostility, contempt, appreciation, or confusion can change how a reply should be interpreted. For example, a reply that appears supportive on the surface may function as a sarcastic challenge in context (Joshi, Bhattacharyya, and Carman 2017; Demszky *et al.* 2020).

Traditional models often treat comments as isolated text, ignoring conversational structure and affective cues. Since context-aware approaches have been shown to improve dialogue act classification (Raheja and Tetreault 2019), this project investigates whether combining dialogue structure, sentiment, and emotion signals improves communicative intent recognition in CMV replies.

We propose a two-stage framework. Stage 1 predicts the dialogue act of each reply using four labels: `agree`, `disagree`, `question`, and `statement`. Stage 2 uses a modular LLM-based pipeline to predict sentiment, emotion, and communicative intent using the reply, its parent comment, and the dialogue act information from Stage 1.

The project constructs a gold-standard subset of 300 short CMV replies sampled from the Winning Arguments Corpus (Cornell Conversational Analysis Toolkit n.d.). The replies are sampled within the 1–30 token range and paired with their parent comments for context. Each reply is annotated for dialogue act, sentiment, emotion, and intent. Emotion labels include `appreciation`, `hostility`, `contempt`, `curiosity_confusion`, `sarcasm`, and `neutral`; intent labels include `Information seeking`, `Challenge`, `Counter-argue`, `Support`, and `Others`.

Through this framework, we examine whether decomposing intent recognition into structural and affective components improves performance over treating replies as isolated text or using a single unified prompt.

Task Description

The goal of this project is to explore whether combining dialogue structure, conversational context, and affective signals improves intent recognition in socially and politically contested online discussions. Using Reddit's r/ChangeMyView (CMV) as the study setting, the project focuses on short replies in argumentative threads. The outcome will be a manually annotated CMV subset with dialogue act, sentiment, emotion, and intent labels, as well as a modular pipeline for intent recognition.

Task 1: Literature Research & State of the Art

This project builds on four areas of prior work: online argumentation, dialogue structure, affective analysis, and modular LLM-based classification. The students are expected to conduct a literature review in these areas and provide a structured overview of key publications, existing techniques, and their connections to the thesis topic.

Task 2: Data Processing and Gold Standard Construction

A gold-standard subset of 300 short CMV replies will be manually annotated by both students. Each reply is paired with its parent comment to preserve conversational context. The samples are annotated along four dimensions: dialogue act, sentiment, emotion, and intent.

To ensure annotation reliability, both annotators will independently label the same 300 samples. Inter-annotator agreement will be measured using Cohen's Kappa, which is commonly used for categorical annotation tasks. A κ value above 0.6 will be considered acceptable agreement.

Disagreements will be resolved through discussion before constructing the final gold-standard evaluation set.

Task 3: Structural Dialogue Act Classification (Stage 1)

The first stage identifies the dialogue act of each reply using a four-label schema: `agree`, `disagree`, `question`, and `statement`. These labels capture the surface function of the reply and provide structural context for interpreting communicative intent. We compare three modelling strategies:

— **Direct LLM Annotation:** LLM-predicted dialogue act labels are used directly without fine-tuning. This establishes whether direct LLM inference is sufficient for dialogue act classification.

— **Silver-Standard Fine-tuning:** An open-source LLM is prompted to annotate dialogue acts on a larger sample of CMV data using zero-shot, few-shot, and chain-of-thought prompting strategies. The resulting silver-standard annotations are used to fine-tune a RoBERTa-based classifier.

— **Gold-Standard Fine-tuning:** RoBERTa is fine-tuned directly on the 300-sample gold-standard annotations using cross-validation. This provides a data-efficiency upper bound against which LLM-based annotation and silver-standard fine-tuning can be compared.

All strategies are evaluated against the 300-sample human-annotated gold standard. The predicted dialogue act labels are then passed as additional input features to Stage 2.

Task 4: Multi-Module Affective Reasoning (Stage 2)

The second stage focuses on extracting affective signals and inferring the communicative intent of short replies. We design a modular LLM-based pipeline and investigate two research questions: (1) whether decomposing the task into separate modules improves performance over a single prompt, and (2) what information-sharing strategy between modules leads to the best intent recognition performance.

Experiment 1: Single-Prompt Baseline vs. Multi-Module Pipeline

We compare two architectures:

— **Single-prompt baseline:** A single LLM call receives the parent comment, the reply, and the dialogue act label from Stage 1, and outputs sentiment, emotion, and intent simultaneously.

— **Multi-module pipeline:** Three independent LLM calls are used, each with a dedicated role:
  - **Sentiment module:** predicts coarse sentiment polarity using three labels: `positive`, `negative`, and `neutral`.
  - **Emotion module:** predicts fine-grained emotional tone using a 6-category taxonomy: `appreciation`, `hostility`, `contempt`, `curiosity_confusion`, `sarcasm`, and `neutral`.
  - **Intent module:** infers communicative intent using five labels: `Information seeking`, `Challenge`, `Counter-argue`, `Support`, and `Others`.

Both architectures are evaluated against the 300-sample gold standard using Macro F1. This experiment tests whether task decomposition into smaller, focused prompts improves label accuracy over a single unified prompt.

Experiment 2: Information-Sharing Strategy Between Modules

We investigate whether passing intermediate predictions between modules improves intent recognition. For intent prediction, we compare the following input combinations:

— reply + parent → intent

— reply + parent + dialogue act → intent

— reply + parent + sentiment → intent

— reply + parent + emotion → intent

— reply + parent + dialogue act + sentiment → intent

— reply + parent + dialogue act + emotion → intent

— reply + parent + dialogue act + emotion + sentiment → intent

Task 5: Evaluation

All tasks are evaluated against the 300-sample gold standard using Macro F1 as the primary metric, with per-class F1 and confusion matrices reported for error analysis.

Stage 1 – Dialogue Act The three modelling strategies are compared using Macro F1. Cohen's Kappa is reported as an additional measure of annotation reliability.

Stage 2 – Emotion, Sentiment, and Intent Emotion is evaluated using Precision, Recall, and Macro F1 across the 6-category binary taxonomy. Sentiment and intent are evaluated using Macro F1. Intent recognition serves as the primary metric for assessing overall system effectiveness.

Experiment 1: Single Prompt vs. Multi-Module Both architectures are compared using Macro F1 for each label type. We report whether task decomposition improves over the single-prompt baseline.

Experiment 2: Information-Sharing Strategy For each input combination, Intent Macro F1 is reported and compared against the text-only baseline. Results identify the optimal module execution order and whether inter-module information sharing is necessary.

Error Analysis Misclassified examples are manually reviewed to identify whether errors originate from incorrect DA predictions, undetected sarcasm, or insufficient conversational context.

Thesis Deliverables

The students must submit complete documentation of the work, including the written report, any generated or annotated data (subject to privacy and license regulations), source code for all components, trained models, and generated results.

Project report The report should use the official template provided by the DDIS supervisor. It should include at least the following sections: 1. Introduction, 2. Related Work, 3. Methodology, 4. Experimental Setup, 5. Results, 6. Discussion and Limitations, and 7. Conclusion.

Alternate structures are possible, but should be discussed and agreed upon with the supervisor.

Research Artifacts All artifacts produced in the course of the thesis must be prepared, documented, and submitted in a manner that enables others to understand and, where applicable, reproduce the work. The specific form this takes depends on the nature of the thesis:

1. Code Base: For theses involving computational work, the code base should be submitted to the DDIS Gitlab repository. The code should be well-structured and contain explanatory comments where appropriate. A README file must provide installation instructions, environment setup details, and clear steps for reproducing the results, including information on dependencies and external resources. Additionally, the codebase should include unit and integration tests for other components, where applicable, to ensure the functionality of key modules outside the training process.
2. Qualitative Study Materials: For theses involving qualitative work (e.g., interviews, user tests, observations), the submission should include study materials such as interview guides, survey instruments, and consent forms, as well as any documentation necessary to understand how the data was collected and analyzed. Raw materials (e.g., transcripts, recordings) should be handled in accordance with applicable privacy and data protection regulations. Consult your supervising PhD student or Postdoc to agree on what can be shared and in what form

Data Documentation Akin to the code base, data should be submitted in a well-documented manner to enable reproduction of the results. Please confer with your supervising PhD student or postdoc to agree on a suitable method that adheres to open science principles and any applicable data-use/-distribution restrictions (e.g., data protection).

Presentation

Upon completion of the project, master's students are required to share the results with the supervising group in a final 15-minute presentation.

General Thesis Guidelines

Please note the following general thesis guidelines

- All source code produced as part of this thesis should be released under a suitable open-source license. Please consult the supervising assistant about how we intend to do so.
- The usual rules of academic work apply; see **Bernstein:2010p446** for general guidance.
- At the end of the thesis, a written report must be submitted. The report should be clearly organized and follow a standard academic structure.
- Any use of AI should be agreed upon with the supervising assistant and needs to be declared according to the UZH, Oec, and IfI rules.
- Grading will follow the template available at `https://www.ifi.uzh.ch/dam/jcr:d21e01dd-fe88-42d4-80b1-342ac4d137ec/grading_overview.pdf`

Signatures

Xin Zhong Xiaying Ji

Xin Zhong & Xiaying Ji

Prof. A. Bernstein, Ph.D.

Note that the signing procedure is the following: The student sends the signed PDF of this task description together with all the signed thesis registration forms as PDFs to `studies@ifi.uzh.ch` with a cc to Prof. Bernstein and, assuming there is one, the supervising Assistant (PhD or Postdoc). Prof. Bernstein's signature is not needed. The student office will contact you with a registration confirmation.

References

Cornell Conversational Analysis Toolkit (n.d.). *Winning Arguments Corpus*. https://convokit.cornell.edu/documentation/winning.html. Accessed through ConvoKit.

Demszky, Dorottya *et al*. (2020). “GoEmotions: A Dataset of Fine-Grained Emotions”. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4040–4054. DOI: 10.18653/v1/2020.acl-main.372. URL: https://aclanthology.org/2020.acl-main.372/.

Joshi, Aditya, Pushpak Bhattacharyya, and Mark J. Carman (2017). “Detecting Sarcasm Using Different Forms of Incongruity”. In: *Proceedings of the 8th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, p. 1. DOI: 10.18653/v1/W17-5201. URL: https://aclanthology.org/W17-5201/.

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Tan, Chenhao *et al*. (2016). “Winning Arguments: Interaction Dynamics and Persuasion Strategies in Good-faith Online Discussions”. In: *Proceedings of the 25th International Conference on World Wide Web*, pp. 613–624. DOI: 10.1145/2872427.2883081. URL: https://arxiv.org/abs/1602.01103.

Appendices

Example To illustrate the annotation scheme, consider the following simplified example:

— *Parent comment:* “Vaccines are dangerous and should be banned.”

— *Reply:* “Sure, great logic.”

The reply is annotated along four dimensions:

— *Dialogue Act:* **disagree**. Although the reply contains agreement-like wording, it rejects the parent comment in context.

— *Sentiment:* **negative**. The reply expresses a negative stance toward the parent comment.

— *Emotion:* **sarcasm**. The phrase “great logic” is ironic rather than sincere praise.

— *Intent:* **challenge**. The reply challenges the reasoning of the parent comment rather than providing a detailed counter-argument.

Final annotation output:

— *Dialogue Act:* disagree

— *Sentiment:* negative

— *Emotion:* sarcasm

— *Intent:* challenge

— *Explanation:* The reply appears positive on the surface, but the context makes the praise ironic. It is therefore annotated as a negative, sarcastic challenge.

This example shows how context helps distinguish literal wording from the intended communicative meaning of a short reply.

Dialogue Act Annotation Dialogue act labels capture the surface function of a reply. Each reply is assigned one dialogue act label.

Label	When to mark	Example
`agree`	Agreement, acceptance, or confirmation of the parent comment.	*“You are correct.”*
`disagree`	Rejection, denial, or disagreement with the parent comment.	*“This makes no sense to me.”*
`question`	A question, clarification request, or information-seeking reply.	*“Can you please define what statutory rape is?”*
`statement`	Information, explanation, elaboration, or assertion without a clear agree/disagree/question form.	*“Cable companies are licensed monopolies and competition is prohibited.”*

Table 1: Dialogue act annotation guideline.

Sentiment Annotation Sentiment labels capture the overall polarity of a reply. Each reply is assigned one sentiment label.

Label	When to mark	Example
`positive`	Positive stance, approval, praise, gratitude, or friendly recognition.	*“Good points.”*
`negative`	Negative stance, criticism, rejection, hostility, sarcasm, or contempt.	*“You should do some basic research before posting here.”*
`neutral`	No clear positive or negative polarity; mainly factual, explanatory, or information-seeking.	*“It would at first until it became common knowledge.”*

Table 2: Sentiment annotation guideline.

Emotion Annotation Emotion labels capture the affective tone of a reply, separate from dialogue act and sentiment. Each label is binary; multiple non-neutral labels may be selected. `neutral` is used when no clear emotional tone is expressed.

Label	When to mark 1	Example
`appreciation`	Gratitude, praise, warmth, or positive recognition. We use this instead of `happy`, since positive replies usually express recognition rather than general happiness.	*"Good points."*
`hostility`	Anger, irritation, impatience, aggression, or hostile confrontation.	*"You should do some basic research before posting here."*
`contempt`	Mockery, belittlement, dismissiveness, or looking down on the speaker or argument.	*"Verbosity isn't the path to being correct."*
`curiosity_confusion`	Curiosity, confusion, uncertainty, lack of understanding, or need for clarification.	*"I'm not sure I'm following you."*
`sarcasm`	Irony, insincere praise, or wording whose intended meaning differs from its literal meaning. May co-occur with `contempt` or `hostility`.	*"Great counterargument. Clearly your logical reasoning..."*
`neutral`	No clear emotional tone; mainly factual, explanatory, or argumentative.	*"It would at first until it became common knowledge."*

Table 3: Emotion annotation guideline.

Intent Annotation Intent labels capture the communicative purpose of a reply in context. Unlike dialogue act, which describes the surface form, intent describes what the reply is trying to accomplish. Each reply is assigned one intent label.

Label	When to mark	Example
`Information seeking`	The reply aims to obtain information, clarification, definition, or explanation.	*"What do you mean by that?"*
`Challenge`	The reply questions, doubts, or pushes back without giving a full counter-argument.	*"But do we have good reason to believe that?"*
`Counter-argue`	The reply provides reasons, evidence, or explanation against the parent comment.	*"Cable companies are licensed monopolies and competition is prohibited."*
`Support`	The reply supports, agrees with, strengthens, or positively acknowledges the parent comment.	*"That latter point is a very good one."*
`Others`	The reply does not clearly fit the above categories, such as jokes, meta-comments, procedural remarks, or unclear responses.	*"Happy 100Δ!"*

Table 4: Intent annotation guideline.